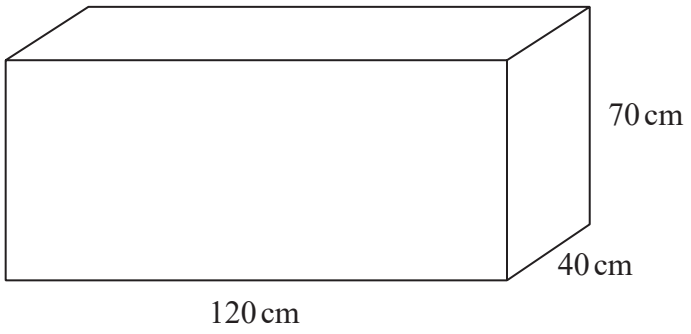
17 The diagram shows Helen's fish tank.
The fish tank is in the shape of a cuboid.

All the dimensions are correct to the nearest centimetre.



Helen is going to use a bucket to fill the fish tank completely with water.
There are 14 litres, correct to the nearest litre, of water in a full bucket.

Will 25 full buckets of water definitely fill the fish tank?
Justify your answer.

Largest dimension bounds: 120.5,40.5,70.5cm.

1 mark
Correct bounds

Tank volume upper bound = 120.5 × 40.5 × 70.5

= 344057.625 cm^3 = 344.057625 litres.

1 mark
Largest tank volume

Least water per bucket = 13.5 litres.

25 buckets can contain only25 × 13.5 = 337.5 litres.

1 mark
Least water in25

No:337.5 < 344.057625, so25 buckets are not guaranteed

to fill a tank close to its largest possible volume.

1 mark
No; compare litres

(Total for Question 17 is 4 marks)